Medium Propagation Campaign A
Open Quantum Transport Laboratory

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Abstract

This report documents the first spatial-medium campaign of the transport laboratory. The campaign replaces purely topological graph comparisons by explicit media with coordinates, spatial spreading observables, and deterministic regime labels. The scanned controls are the disorder strength relative to the coherent coupling and the phase-scrambling rate relative to the same coupling. The output is organized as a reviewed campaign with figures, tables, numerical diagnostics, literature-expectation tracking, and a next-action recommendation.

1 Scope

We study how an initially localized excitation moves through three small open quantum media: a one-dimensional chain, a ring, and a two-dimensional square lattice. The coherent part of the model is a tight-binding Hamiltonian, while the open-system part includes local phase scrambling, target capture, and parasitic loss. The relevant observables are:

- final target population, interpreted as transport success;
- final mean squared displacement, interpreted as spatial spreading;
- final graph-normalized von Neumann entropy, interpreted as internal mixing;
- deterministic regime labels derived from explicit thresholds.

This work follows the general open-system framework of Breuer and Petruccione, the transport perspective emphasized by Plenio and Huelga, the environment-assisted transport picture of Mohseni *et al.*, and graph-transport reviews such as Mülken and Blumen [1, 5, 3, 4, 2].

2 Geometry and maps

3 Automated review summary

A campanha comparou 3 meios e encontrou uma faixa de sucesso de transporte entre 0.228 e 0.803. O espalhamento espacial variou entre 2.882 e 39.948, e a mistura interna variou entre 0.000 e 2.762. Isso ja permite falar em regioes fisicas diferentes do mapa, em vez de so mostrar numeros isolados.

Table 1: Best point found for each medium family in campaign A.

Medium	Best success	Disorder / coupling	Phase scrambling / coupling	Best reg
1D chain medium	0.642	0.00	0.00	coherent-dor
Ring medium	0.803	0.00	0.00	coherent-dor
2D square lattice medium	0.668	0.00	0.00	coherent-dor

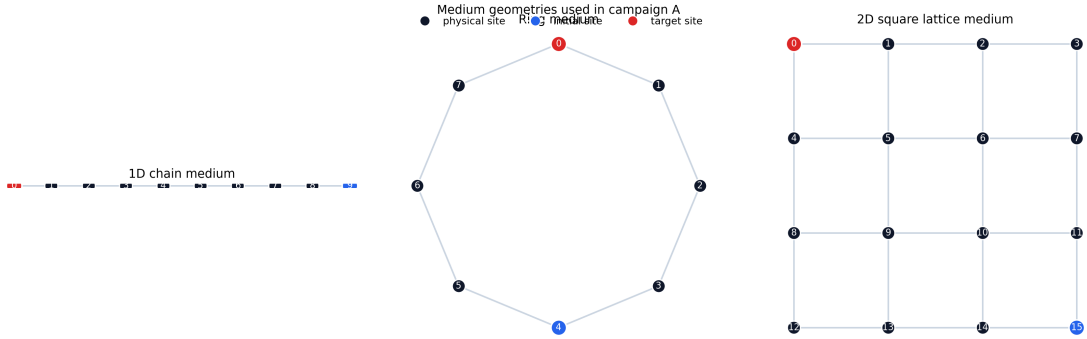

Figure 1: Physical geometries used in campaign A. Blue marks the initial site and red marks the target site that feeds the successful arrival channel.

3.1 Scout findings

Finding: A campanha ja separa chegada ao alvo, espalhamento pelo meio e mistura interna do estado.
Reason: Isso evita confundir boa chegada ao alvo com simples espalhamento dentro do meio.

Finding: Ha regioes de mudanca de regime dentro do mapa.
Reason: Os rotulos calculados nao ficaram todos iguais, entao ha fronteiras fisicas a explorar.

3.2 Critic assessment

Level: ok
Concern: Nenhum problema estrutural imediato apareceu na leitura automatica.
Evidence: As tres familias de observaveis variaram e os regimes nao colapsaram num unico rotulo.

3.3 Planner recommendation

Next action: refine a boundary
Reason: O proximo passo mais seguro e refinar a fronteira do regime dominante observado (mixed-crossover) em vez de abrir um meio novo cedo demais.

4 Scenario-by-scenario reading

4.1 1D chain medium

This medium uses the geometry type chain_1d with 10 physical sites, coherent coupling $J = 1.00$, target-capture rate $\kappa = 0.65$, and parasitic loss rate $\Gamma = 0.02$. The best point found in the

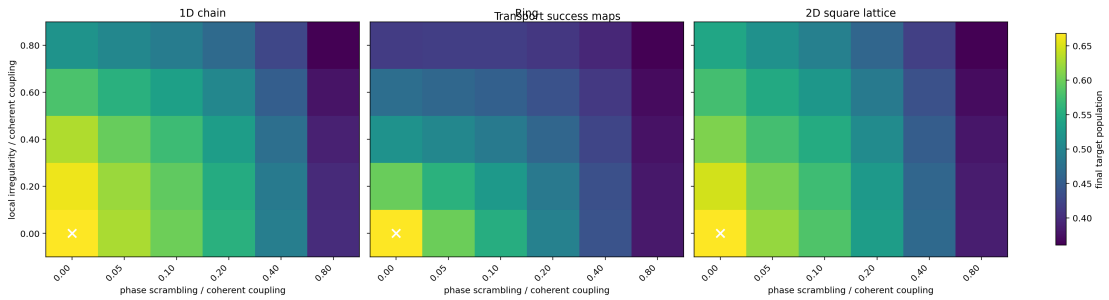

Figure 2: Transport-success maps. Each panel shows the final target population as a function of local irregularity divided by coherent coupling and phase scrambling divided by coherent coupling. White crosses mark the best point found in each medium.

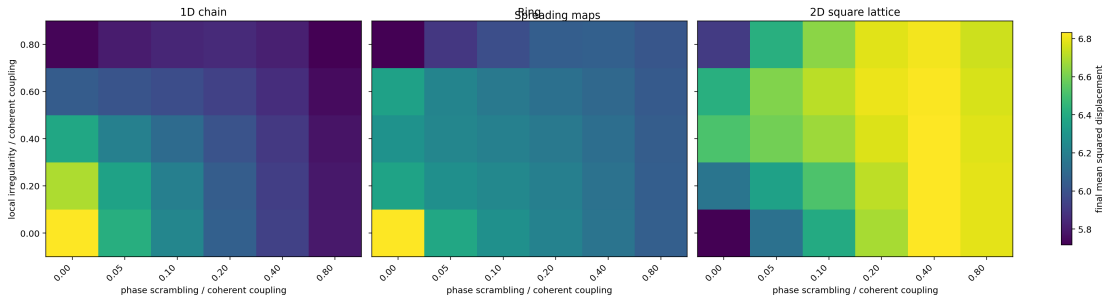

Figure 3: Spreading maps. The plotted quantity is the final mean squared displacement computed from the graph-normalized node populations and the explicit medium coordinates.

scanned map occurs at disorder-to-coupling ratio $W/J = 0.00$ and phase-scrambling-to-coupling ratio $\gamma_\phi/J = 0.00$, giving final success $\eta = 0.642$, final spread 39.948, final mixing entropy 0.000, and regime label coherent-dominated.

Literature expectation. Expected transport trend: Weak disorder should preserve coherent transport, while stronger disorder can be partially mitigated by moderate phase scrambling. Expected role of disorder: Increasing local irregularity should reduce spreading and lower arrival at the target. Expected role of phase scrambling: Moderate phase scrambling may help once coherent motion begins to get trapped by disorder. Expected failure mode: Too much phase scrambling should suppress spreading and reduce transport.

Measured comparison. Disorder suppression observed: True. Moderate phase scrambling helps: False. Strong phase scrambling suppresses transport: True. Agreement level: high with score 0.67.

4.2 Ring medium

This medium uses the geometry type ring with 8 physical sites, coherent coupling $J = 1.00$, target-capture rate $\kappa = 0.65$, and parasitic loss rate $\Gamma = 0.02$. The best point found in the scanned map occurs at disorder-to-coupling ratio $W/J = 0.00$ and phase-scrambling-to-coupling ratio $\gamma_\phi/J = 0.00$, giving final success $\eta = 0.803$, final spread 3.904, final mixing entropy 0.000, and regime label coherent-dominated.

Literature expectation. Expected transport trend: Symmetry can help or hinder transport depending on target geometry, so ring transport may benefit from moderate phase scrambling. Expected role of disorder: Disorder should break perfect symmetry and can either help or hinder

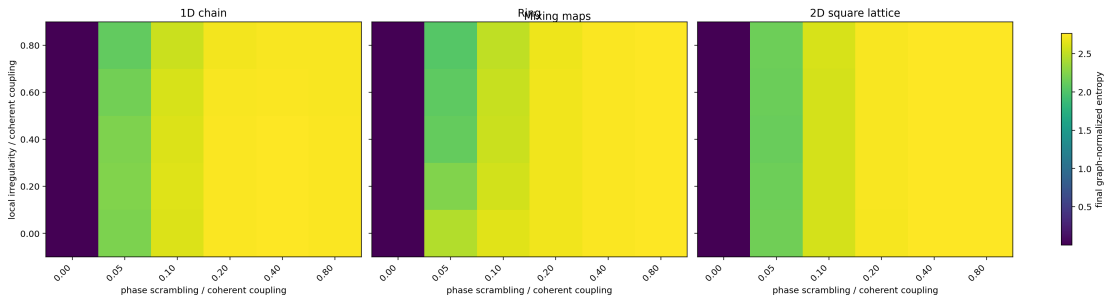

Figure 4: Mixing maps. The plotted quantity is the final graph-normalized von Neumann entropy, used here as a measure of internal mixing in the physical sites.

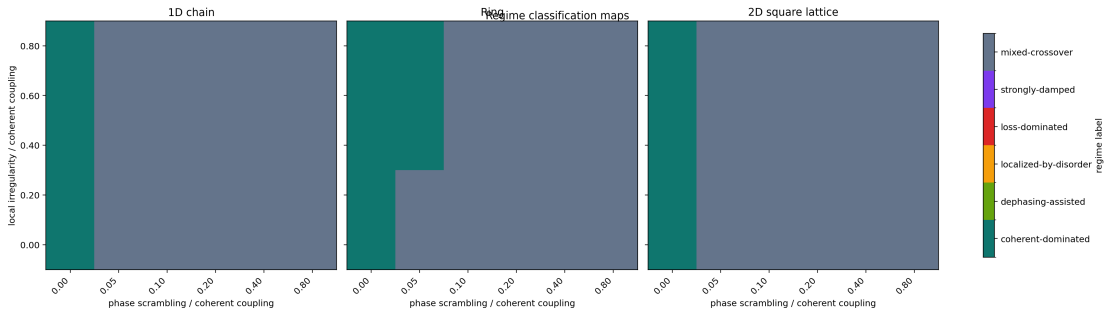

Figure 5: Deterministic regime-label maps generated from explicit thresholds rather than free-form text.

depending on the baseline interference pattern. Expected role of phase scrambling: A moderate amount can break destructive interference and improve arrival at the target. Expected failure mode: Very strong phase scrambling should wash out coherent structure without preserving efficient long-range motion.

Measured comparison. Disorder suppression observed: True. Moderate phase scrambling helps: False. Strong phase scrambling suppresses transport: True. Agreement level: high with score 0.67.

4.3 2D square lattice medium

This medium uses the geometry type square_lattice_2d with 16 physical sites, coherent coupling $J = 1.00$, target-capture rate $\kappa = 0.65$, and parasitic loss rate $\Gamma = 0.02$. The best point found in the scanned map occurs at disorder-to-coupling ratio $W/J = 0.00$ and phase-scrambling-to-coupling ratio $\gamma_\phi/J = 0.00$, giving final success $\eta = 0.668$, final spread 5.717, final mixing entropy 0.000, and regime label coherent-dominated.

Literature expectation. Expected transport trend: A 2D medium should offer more alternate paths, so transport can remain viable even when one path is degraded. Expected role of disorder: Stronger irregularity should reduce clean spreading and may create trapped regions. Expected role of phase scrambling: A moderate amount may stabilize target arrival by reducing coherent path competition. Expected failure mode: Excessive phase scrambling should turn the motion into slow, heavily damped spreading.

Measured comparison. Disorder suppression observed: True. Moderate phase scrambling helps: False. Strong phase scrambling suppresses transport: True. Agreement level: high with score 0.67.

5 Interpretive note

The report is intentionally conservative. A regime label is treated as meaningful only when it is supported by the transport map, the spreading map, and the mixing map simultaneously. The purpose of this campaign is not to make the strongest possible claim yet, but to establish a spatial baseline that the next campaigns can refine and extend.

References

[1] Heinz-Peter Breuer and Francesco Petruccione. *The Theory of Open Quantum Systems*. Oxford University Press, 2002.

[2] Filippo Caruso, Alex W. Chin, Animesh Datta, Susana F. Huelga, and Martin B. Plenio. Highly efficient energy excitation transfer in light-harvesting complexes: the fundamental role of noise-assisted transport. *The Journal of Chemical Physics*, 131(10):105106, 2009.

[3] Masoud Mohseni, Patrick Rebentrost, Seth Lloyd, and Alán Aspuru-Guzik. Environment-assisted quantum walks in photosynthetic energy transfer. *The Journal of Chemical Physics*, 129(17):174106, 2008.

[4] Oliver Mülken and Alexander Blumen. Continuous-time quantum walks: models for coherent transport on complex networks. *Physics Reports*, 502(2-3):37–87, 2011.

[5] Martin B. Plenio and Susana F. Huelga. Dephasing-assisted transport: quantum networks and biomolecules. *New Journal of Physics*, 10(11):113019, 2008.